

Dealer Ops

Dealer Management, CRM & OMVIC Ad-Compliance Platform

Technical & Business Specification · Student Project Brief

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Prepared for classroom use — a worked example of turning a client brief into a technical spec.

1. Overview

Dealer Ops is a multi-tenant web application for independent car dealers, in the spirit of platforms like AutoManager. A single platform admin provisions each dealer business and issues login credentials; a dealer may hold several credentials (one per staff member), all of which share the same underlying data for that dealer.

The system has three connected "upstream" modules — Dealer Management (DMS), Customer Relationship Management (CRM), and Ad Compliance — each with its own front-end entry form, all writing to a shared back-end data store rather than existing as separate, disconnected tools.

2. Roles & Access

Role	Can do	Cannot do
Platform admin	Create dealer businesses. Create, view and remove dealer login credentials.	View or edit any dealer's DMS, CRM or ad data.
Dealer user	Full read/write on their dealer's DMS, CRM and ad-compliance records. Multiple users can share one dealer.	See other dealers' data. Create or remove logins.

Every dealer user signs in with their own username and password; the account record links back to exactly one dealer, and every write from that account is scoped to that dealer's data only.

3. Dealer Management System (DMS)

One record per vehicle, tracked from acquisition through sale.

Required fields

- **Make***
- **Model***
- **Year***
- **VIN***
- **Car source*** — trade-in, auction, private purchase, etc.
- **Purchase cost***
- **Date added to garage***
- **Vehicle condition*** — Certified, As-is, Unfit, or Irreparable

Optional fields

- **Repair cost**
- **Link to Carfax report**
- **Date car was sold**
- **Price car was sold for**

4. Customer Relationship Management (CRM)

One record per customer, with purchases linked back to DMS vehicle records.

Required fields

- **Name***
- **Email***
- **Phone***
- **Home address***
- **Car(s) purchased** — linked selection of that dealer's DMS records

5. Ad Compliance Checker

Before a listing is published, the salesperson completes a checklist and the system flags anything OMVIC requires that is still missing.

Always required

- **Dealer's registered name and contact info**
- **Previous-use disclosure, if applicable** — used as a police/emergency vehicle, taxi/limousine, or daily lease/rental
- **Status (used/new) and year**
- **Extended warranty terms, if offered**
- **Price**
- **Vehicle condition** — Certified, As-is, Unfit, or Irreparable

If the ad shows a rate or payment (financing/credit ads)

- **APR** — shown as prominently as the payment
- **Loan term**
- **Cash price / cost of borrowing**

Exception: radio, TV and billboard ads are exempt from displaying these alongside the rate.

Lease ads must disclose

- **That it is a lease**
- **Term**
- **Payment amount**
- **APR**
- **Upfront payment amount**
- **Excess-km cost** — required only when the annual allowance is under 20,000 km/year

6. Cross-Cutting Requirements

- **Audit trail:** Each modification to a DMS or CRM record is stamped with the acting user, the action taken, and a timestamp, and kept as that record's change history.

- **Architecture:** DMS, CRM and Ad Compliance are plain front ends over one shared back end per dealer — a form for data entry, a store for persistence — not three separate applications.
- **Multi-tenancy:** One dealer, many logins, one shared dataset; the admin has no visibility into dealer data.

7. Pages to Build

Page	Access	Purpose
Login	Public	Dealer and admin sign-in.
DMS	Logged-in dealer	Add, edit and review vehicle inventory.
CRM	Logged-in dealer	Add, edit and review customers; link purchases.
Ad compliance	Logged-in dealer	Check a listing against OMVIC requirements before publishing.
Admin	Platform admin	Create dealers; create and remove login credentials.

8. Known Limitations — For Class Discussion

The working prototype that accompanies this spec makes a few deliberate shortcuts for teaching purposes. Good discussion points for a production build:

- **Authentication:** The prototype stores passwords as plain demo data for readability. A production system must hash and salt passwords server-side and never expose credentials to client code.
- **Compliance:** The checklist encodes OMVIC's current published disclosure rules as reference material, not legal advice — a production system should confirm current requirements with OMVIC and allow the rules to be updated without a code change.
- **Hosting:** The prototype runs inside Claude's artifact hosting for demonstration. A real deployment needs its own domain, hosting, database, and backup strategy.